



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Eleftherios Ntovoris
20 April 2022



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies**

- Data Collection
- Data Wrangling
- EDA with Data Visualization
- EDA with SQL
- Interactive Map with Folium
- Dashboard with Plotly Dash
- Predictive Analysis (Classification)

- **Summary of all results**

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Introduction

- Project background and context
 - The commercial space age is here, companies are making space travel affordable for everyone. Virgin Galactic is providing suborbital space flights. Rocket Lab is a small satellite provider. Blue Origin manufactures sub-orbital and orbital reusable rockets. Perhaps the most successful is SpaceX.
 - One reason it's success is that the rocket launches are relatively inexpensive. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upwards of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage.
 - Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. Instead of using deterministic models we will rely on data.
- Problems you want to find answers
 - Determine the price of each launch
 - Train a machine learning model and use public information to predict if SpaceX will reuse the first stage



Section 1

Methodology

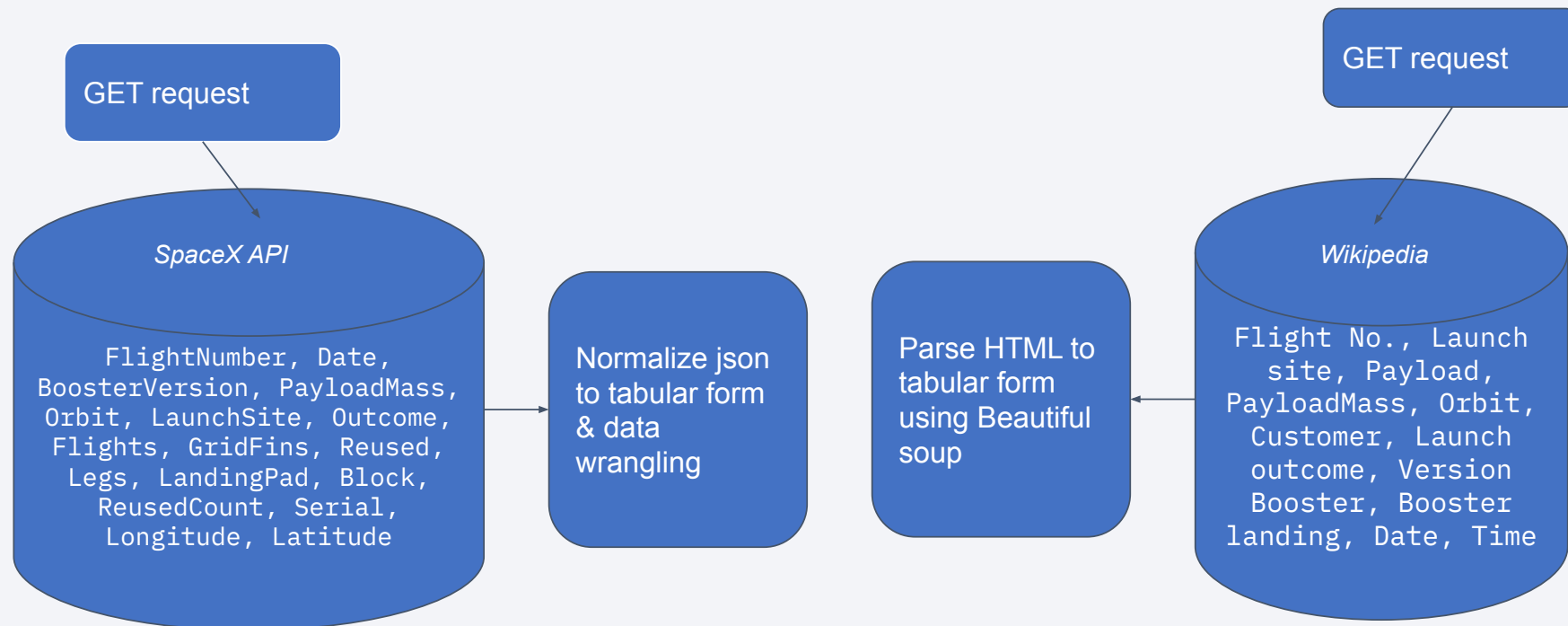
Methodology

Executive Summary

- Data collection methodology:
 - SpaceX REST API and web scraping Wiki for Falcon 9 Launch data
- Perform data wrangling
 - Data cleansing and conversion of the success/failure of landing to binary
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Use cross validation and find optimal hyperparameters for Logistic Regression, Support Vector machines, Decision Tree Classifier, and K-nearest neighbors

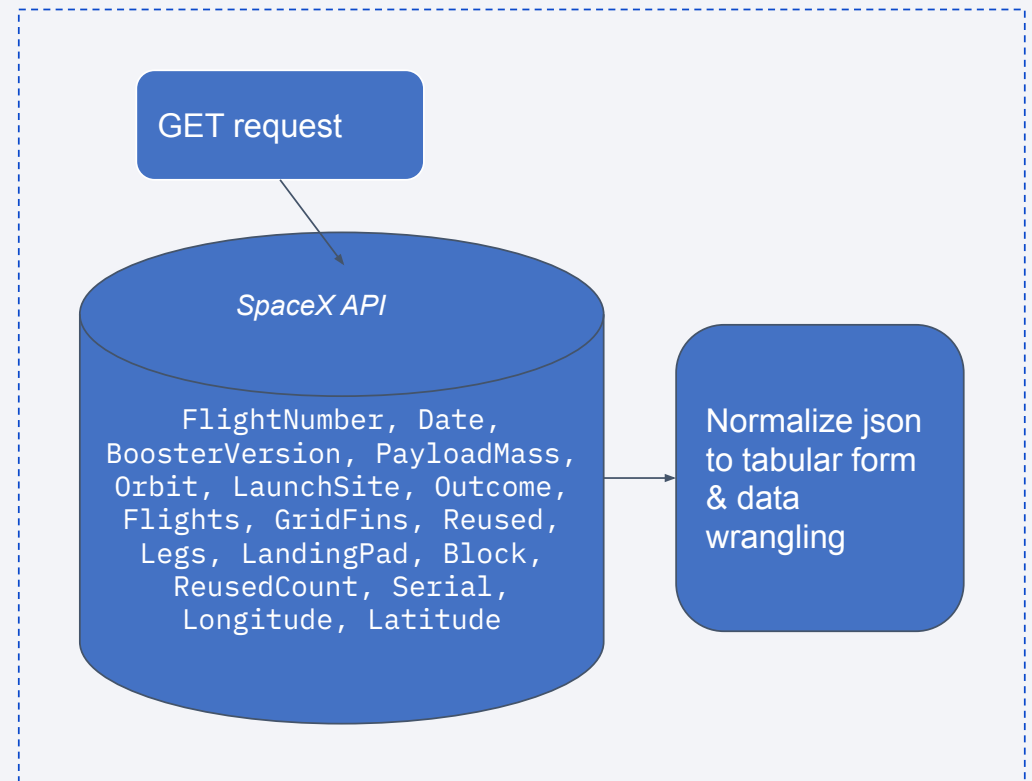
Data Collection

- The data sets were collected from the *SpaceX REST API* and web scraping Wiki pages related to Falcon 9 Launch data.



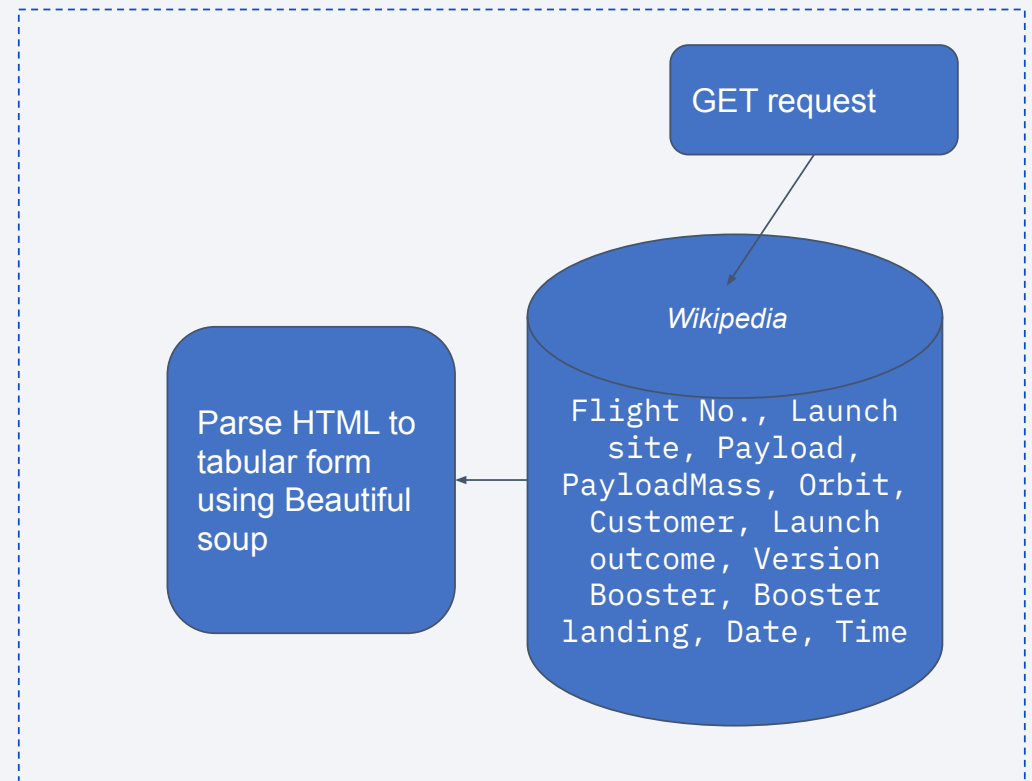
Data Collection – SpaceX API

- Request and parse the SpaceX launch data using the GET request to “<https://api.spacexdata.com/v4/rockets/>” Filter the dataframe to only include Falcon 9 launches. Clean data.
- [GitHub URL](#)



Data Collection - Scraping

- HTTP GET request to the Falcon9 Launch Wiki page from its URL. Extract all column/variable names from the HTML table header. Create a data frame by parsing the launch HTML tables
- [GitHub URL](#)



Data Wrangling

- Missing Values: replace missing values in PayloadMass by the mean
- Explore data for missing values and patterns calculate launches per site and orbit
- Convert outcomes of landing into Training Labels with 1 means the booster successfully landed 0 means it was unsuccessful. The success rate is **67%**

Outcome	Number of occurrences
True ASDS	41
None None	19
True RTLS	14
False ASDS	6
True Ocean	5
False Ocean	2
None ASDS	2
False RTLS	1

True Ocean: mission outcome was successfully landed to the ocean
False Ocean: mission outcome was unsuccessfully landed to the ocean.
True RTLS: mission outcome was successfully landed to a ground pad
False RTLS: mission outcome was unsuccessfully landed to a ground pad
True ASDS: mission outcome was successfully landed to a drone ship
False ASDS: mission outcome was unsuccessfully landed to a drone ship
None ASDS and None None: represent a failure to land

- [GitHub URL](#)

EDA with Data Visualization

- Scatter Plot (Shows correlation between variables)
 - Flight Number vs. Launch Site
 - Payload vs. Launch Site
 - Flight number vs. Orbit type
 - Payload vs. orbit type
- Line Chart
 - Yearly average success rate. To get the average launch success trend
- Bar Chart
 - Success rate of each orbit type. Shows the relationship between success rate and the categorical orbit type
- [GitHub URL](#)

EDA with SQL

- Querying the modeling dataset on a Db2 database on IBM cloud. Queries answered:
 - The names of the unique launch sites in the space mission
 - 5 records where launch sites begin with the string 'CCA'
 - The total payload mass carried by boosters launched by NASA (CRS)
 - Average payload mass carried by booster version F9 v1.1
 - The date when the first successful landing outcome in ground pad was achieved
 - The names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
 - The total number of successful and failure mission outcomes
 - The names of the booster versions which have carried the maximum payload mass
 - The failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015
 - Ranking the count of landing outcomes between 2010-06-04 and 2017-03-20
- [GitHub URL](#)

Build an Interactive Map with Folium

- Map objects that were placed on the map:
 - Markers for each launch site on the site map
 - A marker cluster which shows the success/failure of the landing with color
 - Markers for proximity POIs (places of interest) and lines showing their distance from the launching sites
- The markers help to see the proximity of the markers to the Equator and the coast, any potential correlation of success rate with the geolocation of the launch sites as well as the proximity of the launch sites with railways, highways and major cities.
- [GitHub URL](#)

Build a Dashboard with Plotly Dash

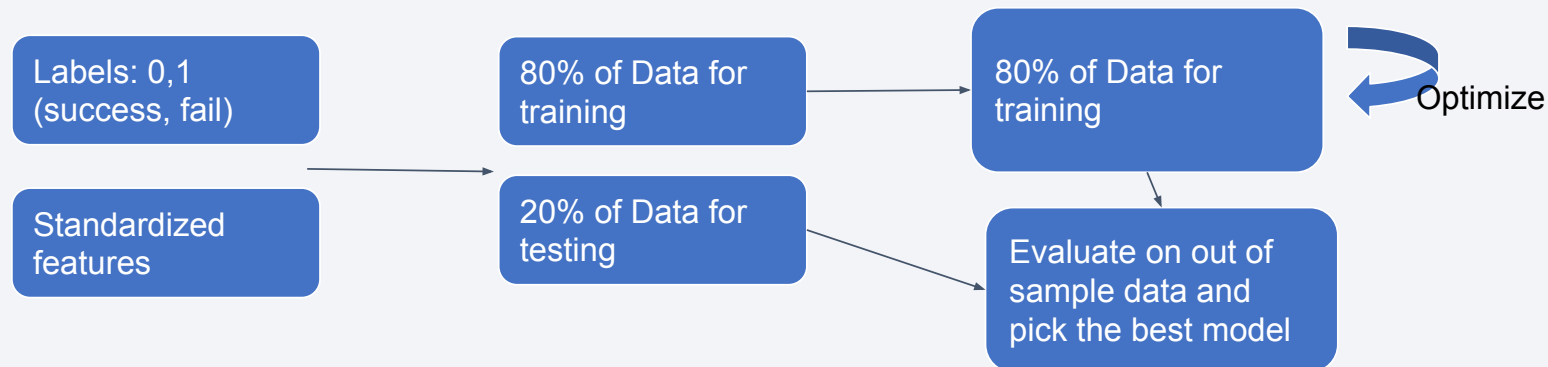
- There are two main interactive plots in the dashboard
 - The Pie chart, which shows the proportion of the successful launches by launch site and can be filtered by a drop down list of all the sites and a slider with the payload range
 - The Scatter plot which show the correlation of the payload mass with the success rate for the different booster categories. The plot is again filtered by the dropdown list of the launch sites and the payload slider

The interactive dashboard gives insights to **which site is more successful** and **which has the largest successful launches**, **which payloads have most/least successful launches** and **which F9 Booster is the most successful**

- [GitHub URL](#)

Predictive Analysis (Classification)

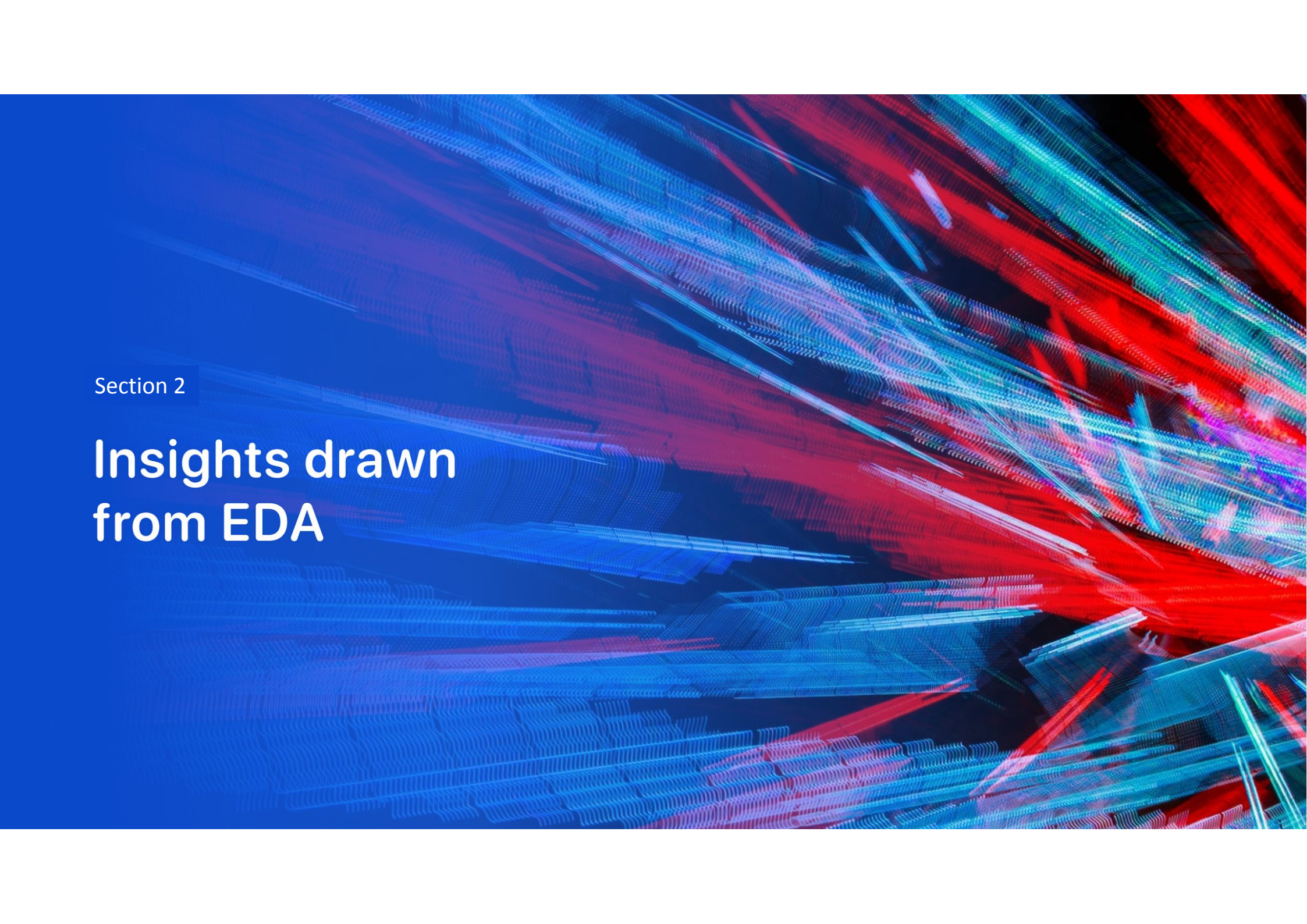
- The success/fail (0 or 1) is the dependent variable to be predicted
- Standardize the independent variables
- Split the modeling set to training (80%) and testing set (20%)
- Fine tune Logistic Regression, Support Vector machines, Decision Tree Classifier, and K-nearest neighbors classifiers by optimizing their hyperparameters using cross validation
- Report the best out of sample (testing set) accuracy score



- [GitHub URL](#)

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

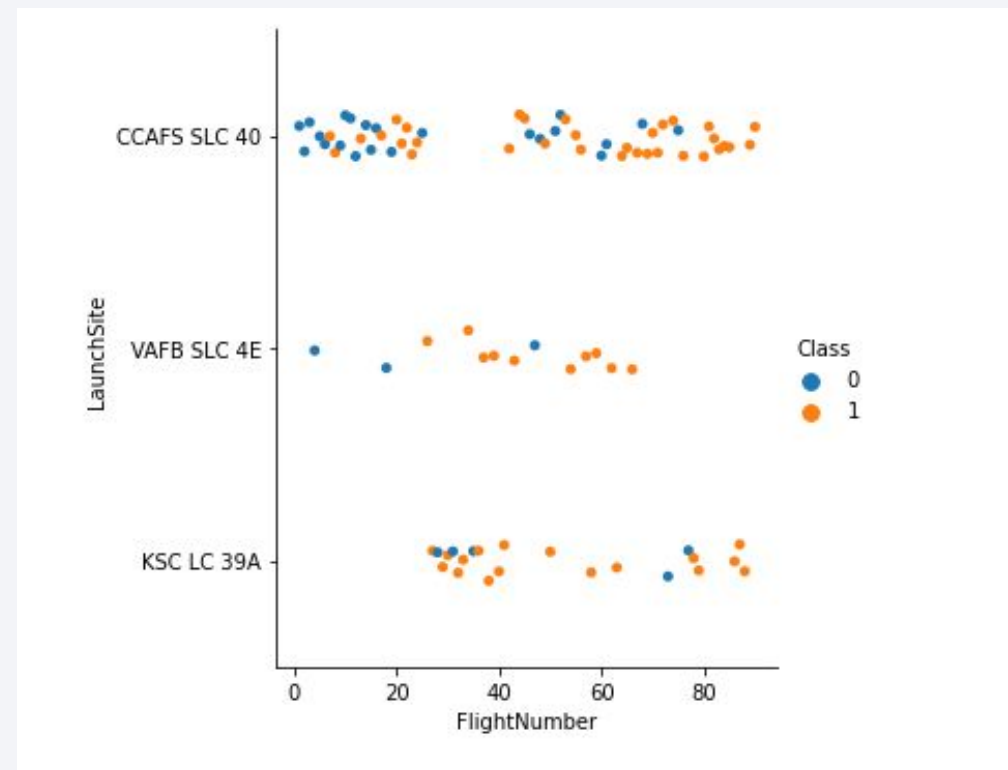


Section 2

Insights drawn from EDA

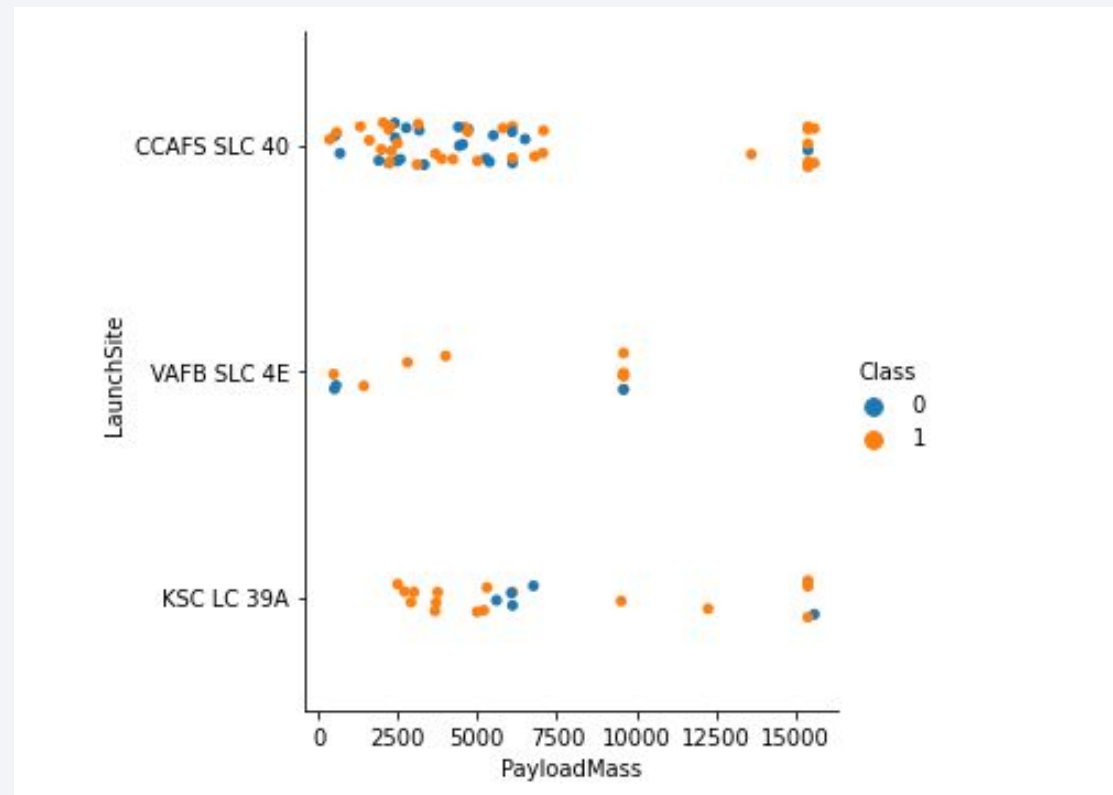
Flight Number vs. Launch Site

- **Class 0**: unsuccessful launch. **Class 1**: successful launch
- For all the launch sites the first flights were less successful
- There are gaps in the launching numbers which might be because of the geolocation of the sites



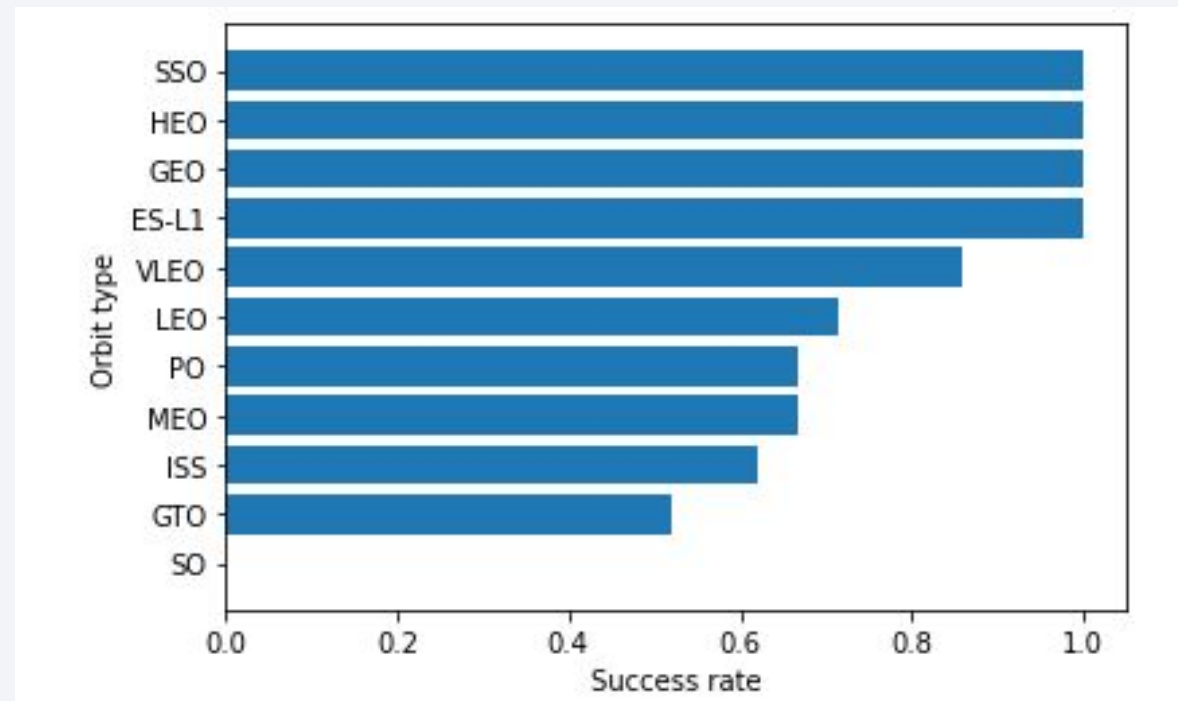
Payload vs. Launch Site

- **Class 0**: unsuccessful launch. **Class 1**: successful launch
- The site VAFB-SLC doesn't launch rockets with heavy payloads
- The site CCAFS SLC might also have some restriction on the payload



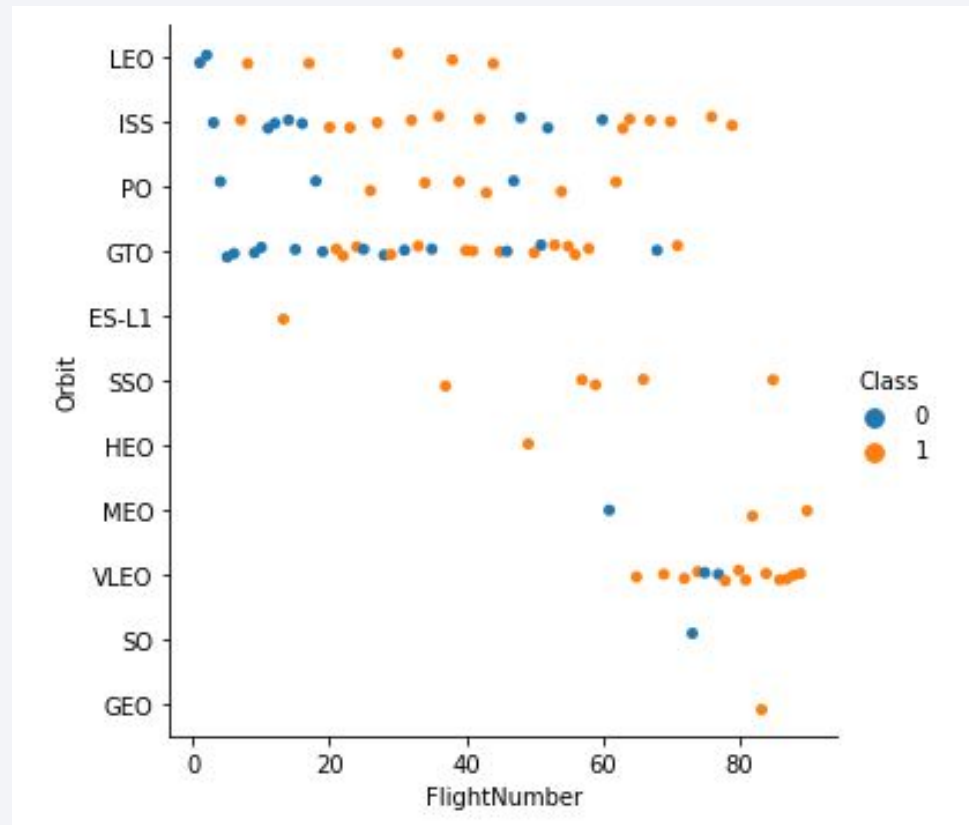
Success Rate vs. Orbit Type

- The orbit types **SSO** (*Sun-synchronous orbit*), **HEO** (*highly elliptical orbit*), **GEO** (*circular geosynchronous*) and **ES_L1** (*earth-sun equilibrium*) have **100%** success rate
- The orbit **SO** (*Sun-synchronous orbit*) has not successful landings
- More than half the orbit types have success rate over **80%**



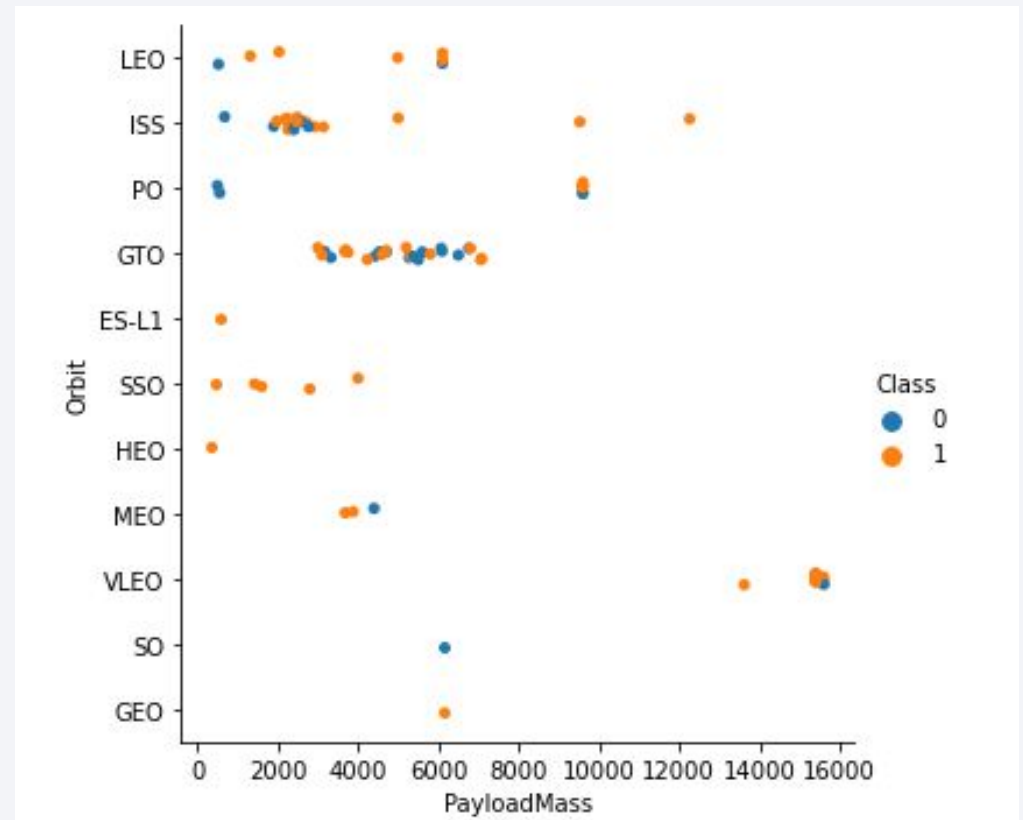
Flight Number vs. Orbit Type

- **Class 0**: unsuccessful launch. **Class 1**: successful launch
- Certain orbits appears to be related with the flight number eg. **LEO**. For some others however this is not true eg. **GTO**
- Some orbits only started to launch later on



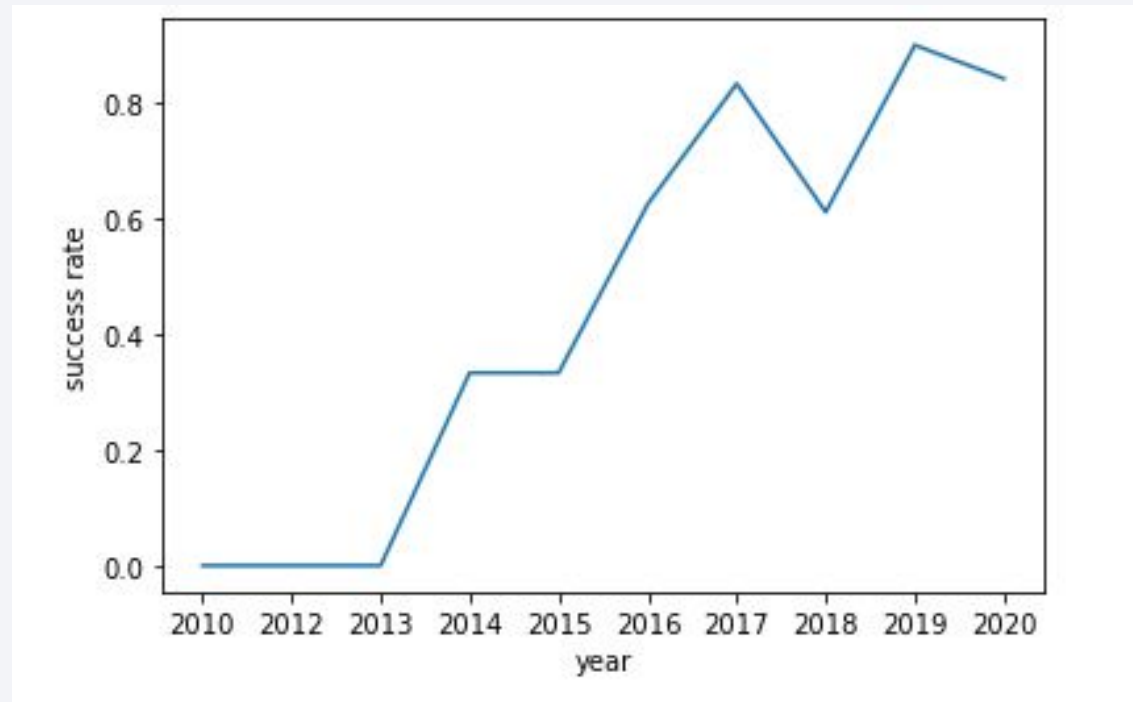
Payload vs. Orbit Type

- **Class 0**: unsuccessful launch. **Class 1**: successful launch
- Polar orbits such as **LEO** and **ISS** appear to do well for heavier payloads
- There is not such distinction for the **GTO** orbit.



Launch Success Yearly Trend

- There is a steady increase in the success rate of the launches since 2013
- The most recent success rate is around 80%
- There was temporary drop on 2018



All Launch Site Names

- Query:

```
select distinct launch_site from  
SPACEXDATASET
```

- Result: There are 4 unique launch sites
- Explanation: Distinct returns unique entries

launch_site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Launch Site Names Begin with 'CCA'

- Query:

```
select * from SPACEXDATASET
where launch_site like 'CCA%'
limit 5
```

Explanation: “%” is a wildcard at the end

- Result:

DATE	time__utc_	booster_version	launch_site	payload	payload_mass__kg_	orbit	customer	mission_outcome	landing__outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Query:

```
select sum(payload_mass__kg_) as total_mass from  
SPACEXDATASET  
where customer = 'NASA (CRS)'
```

- Result: 45596

- Explanation: total payload mass carried by boosters launched by NASA (CRS)

Average Payload Mass by F9 v1.1

- Query:

```
select avg(payload_mass__kg_) as avg_mass from SPACEXDATASET  
where booster_version = 'F9 v1.1'
```

- Result: 2928
- Explanation: the average payload mass carried by booster version F9 v1.1

First Successful Ground Landing Date

- Query:

```
select min(date) as first_success from SPACEXDATASET  
where landing__outcome = 'Success (ground pad)'
```

- Result: 2015-12-22
- The date of the first successful landing outcome on ground pad

Successful Drone Ship Landing with Payload between 4000 and 6000

- Query:

```
select booster_version from SPACEXDATASET
where (payload_mass__kg_ between 4000 and 6000)
and
landing__outcome = 'Success (drone ship)'
```

- The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

- Result:

booster_version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Query:

```
select mission_outcome, count(*) as totals from SPACEXDATASET  
group by mission_outcome
```

Calculates the total number of successful and failure mission outcomes

- Result:

mission_outcome	totals
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Query:

```
select distinct booster_version, payload_mass__kg_ from
(select booster_version, payload_mass__kg_ from SPACEXDATASET
where payload_mass__kg_ = (select max(payload_mass__kg_) from SPACEXDATASET))
```

- Lists the names of the booster which have carried the maximum payload mass

- Result:

booster_version	payload_mass__kg_
F9 B5 B1048.4	15600
F9 B5 B1048.5	15600
F9 B5 B1049.4	15600
F9 B5 B1049.5	15600
F9 B5 B1049.7	15600
F9 B5 B1051.3	15600
F9 B5 B1051.4	15600
F9 B5 B1051.6	15600
F9 B5 B1056.4	15600
F9 B5 B1058.3	15600
F9 B5 B1060.2	15600
F9 B5 B1060.3	15600

2015 Launch Records

- Query:

```
select booster_version, landing__outcome, launch_site from SPACEXDATASET  
where landing__outcome = 'Failure (drone ship)' and year(date) = '2015'
```

- Lists the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

- Result:

booster_version	landing__outcome	launch_site
F9 v1.1 B1012	Failure (drone ship)	CCAFS LC-40
F9 v1.1 B1015	Failure (drone ship)	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Query:

```
select landing__outcome, count(*) as totals from SPACEXDATASET
where date between '2010-06-04' and '2017-03-20'
group by landing__outcome
order by totals desc
```

- Ranks the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

landing__outcome	totals
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

- Result:

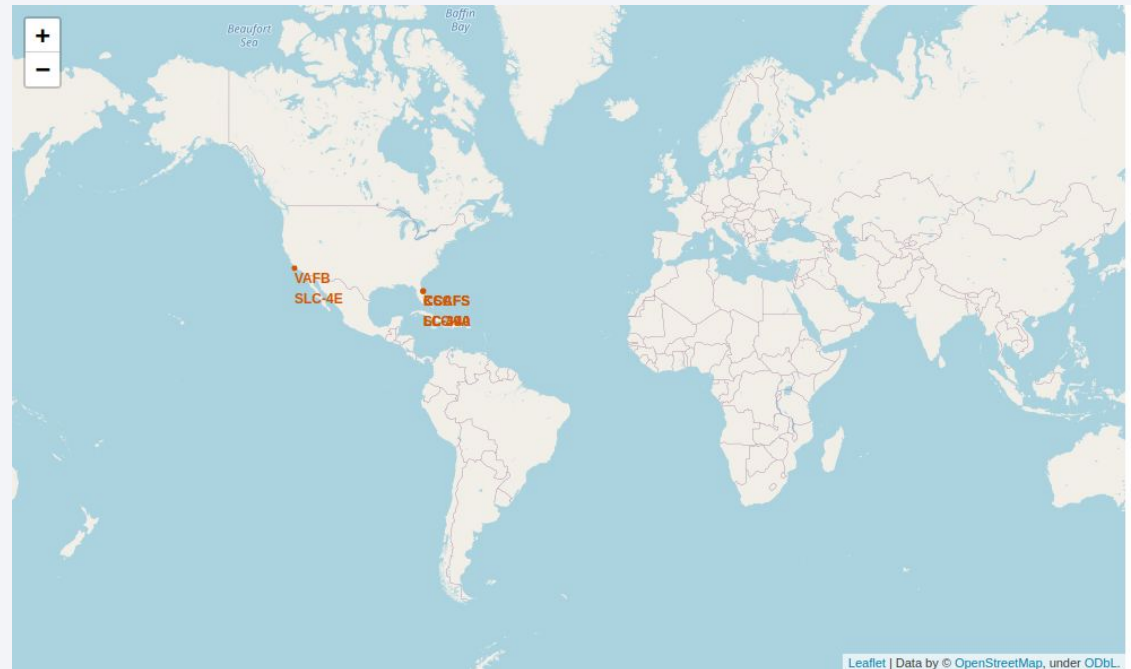
A satellite view of Earth from space, showing the curvature of the planet and the glow of city lights at night. The image is used as a background for the title slide.

Section 3

Launch Sites Proximities Analysis

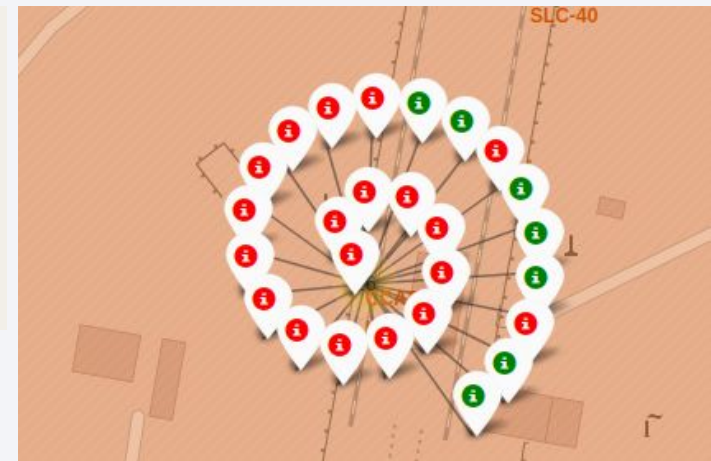
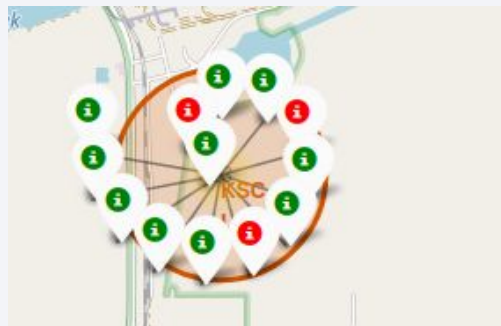
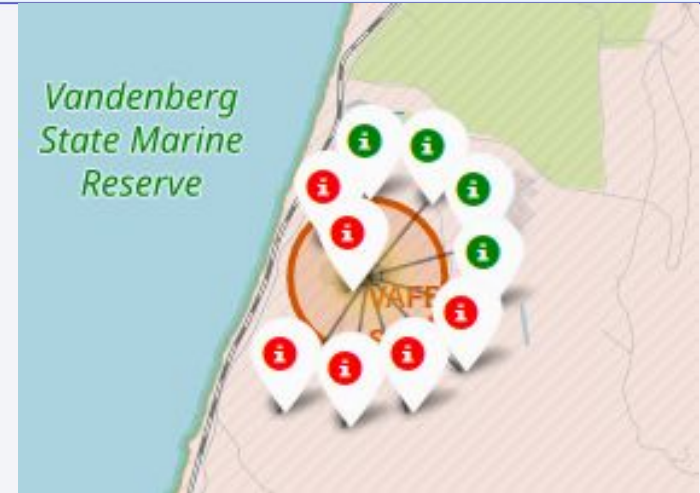
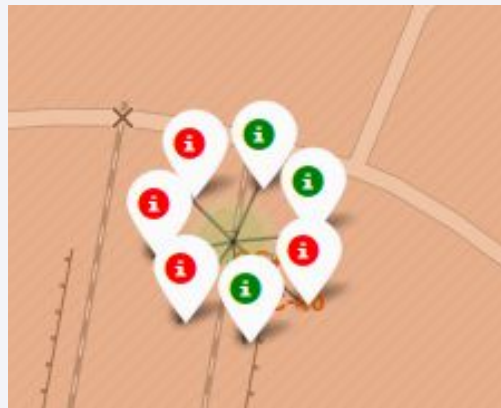
All Launch Sites

- The launch sites are as close as possible to the equator (The launch sites have to be in the United States)
- All the launch sites are near coastal areas



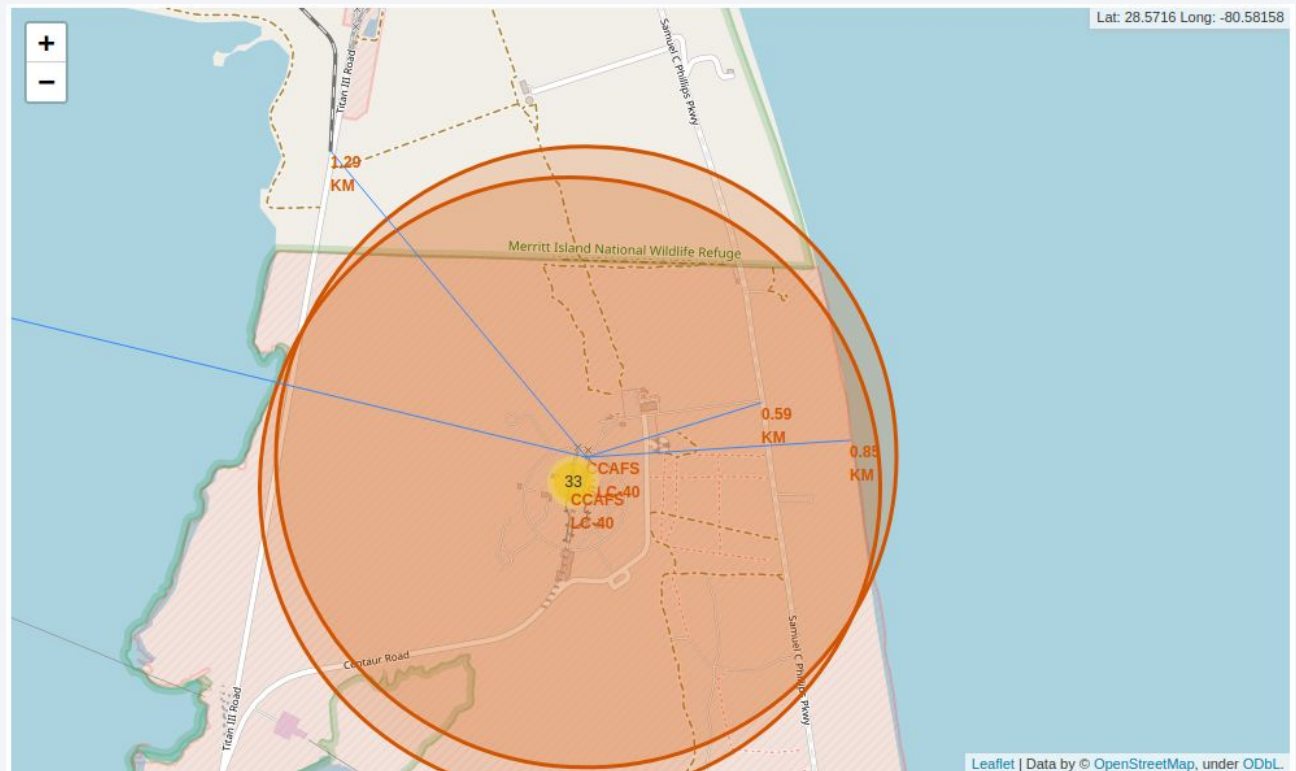
Geolocation vs Success rate

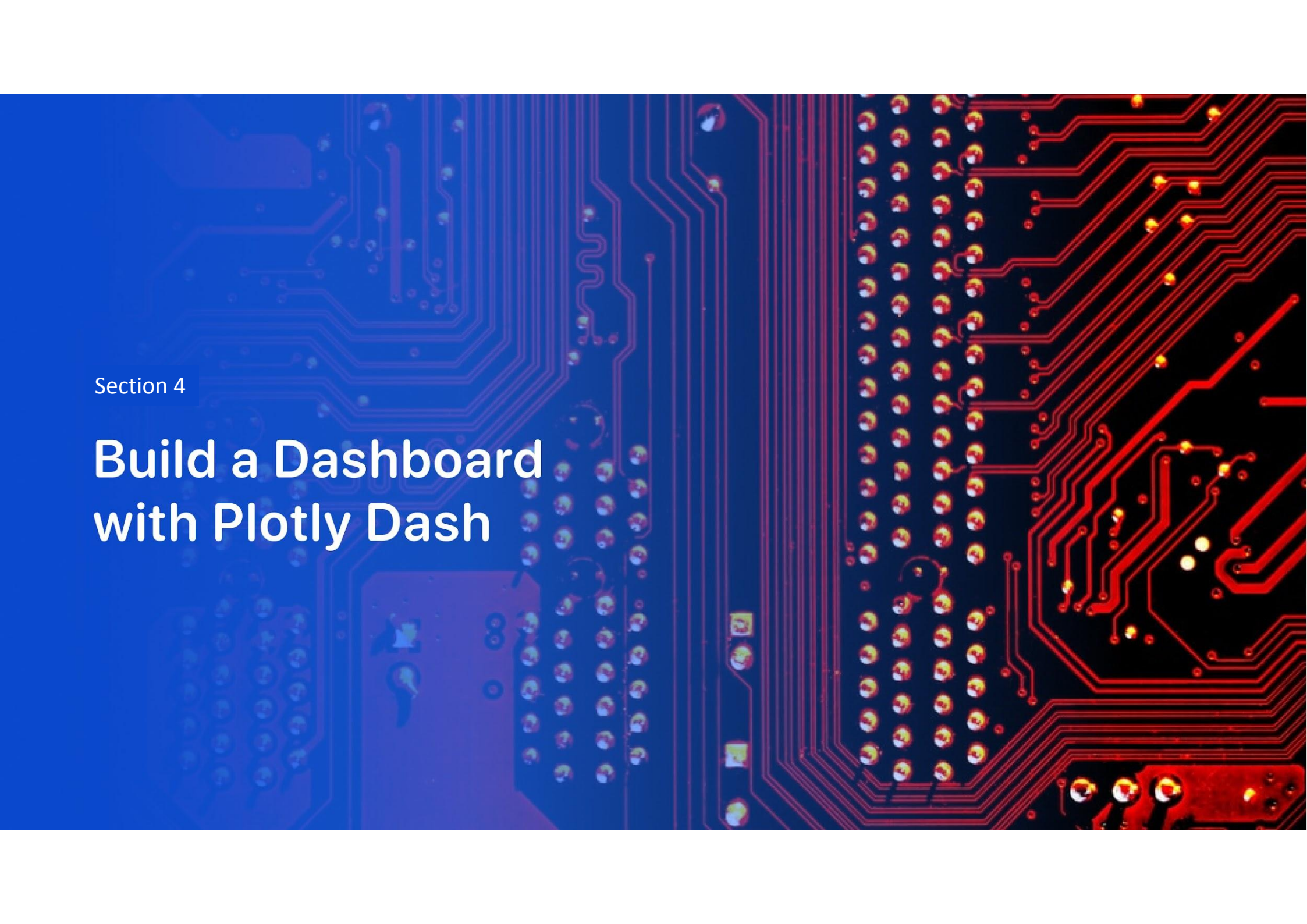
- The red markers are the failed launches while the green are the successful
- The eastern sites KSC LC-39A have higher success rates



Proximities to Launch Sites

Launch site are close to railways, the highway and the coastal line but are far from big city centers



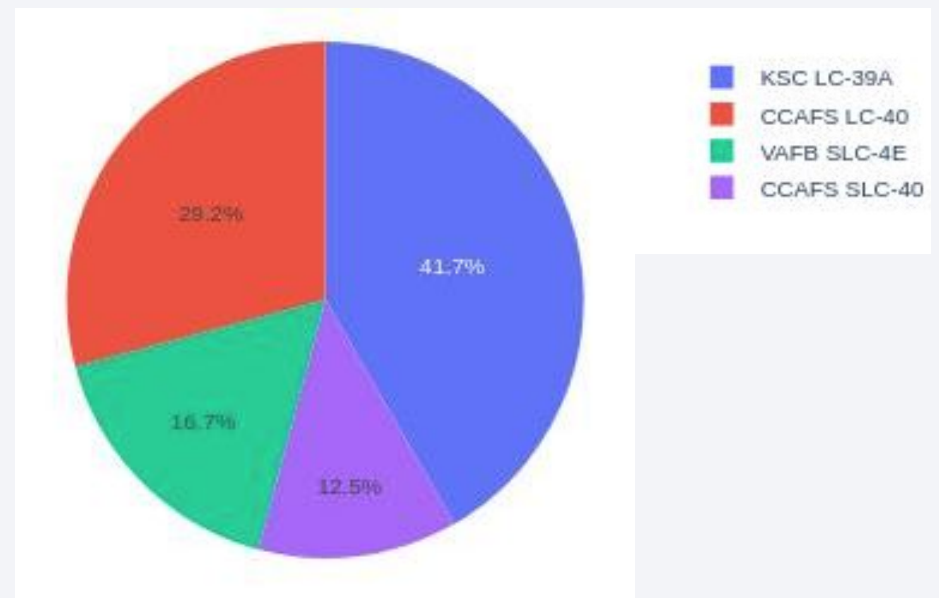


Section 4

Build a Dashboard with Plotly Dash

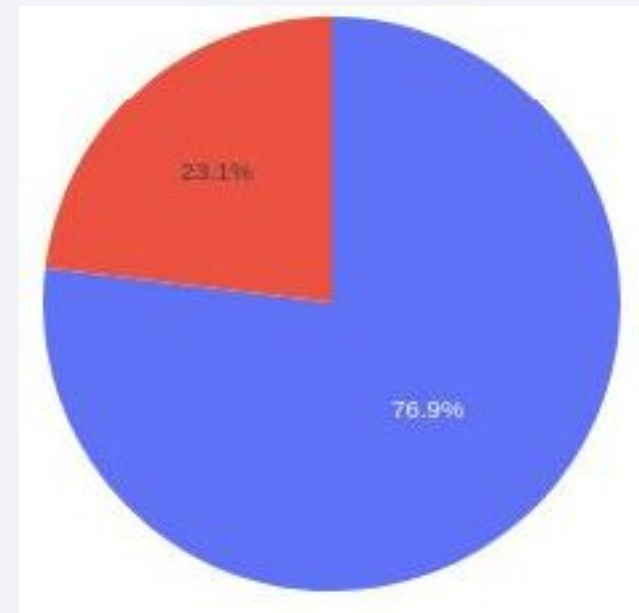
Successes (%) by Launch Site

- KSC LC-39A has the biggest success rate as we have already seen on the map
- The least successful site is CCAFS SLC-40, but there are many launches



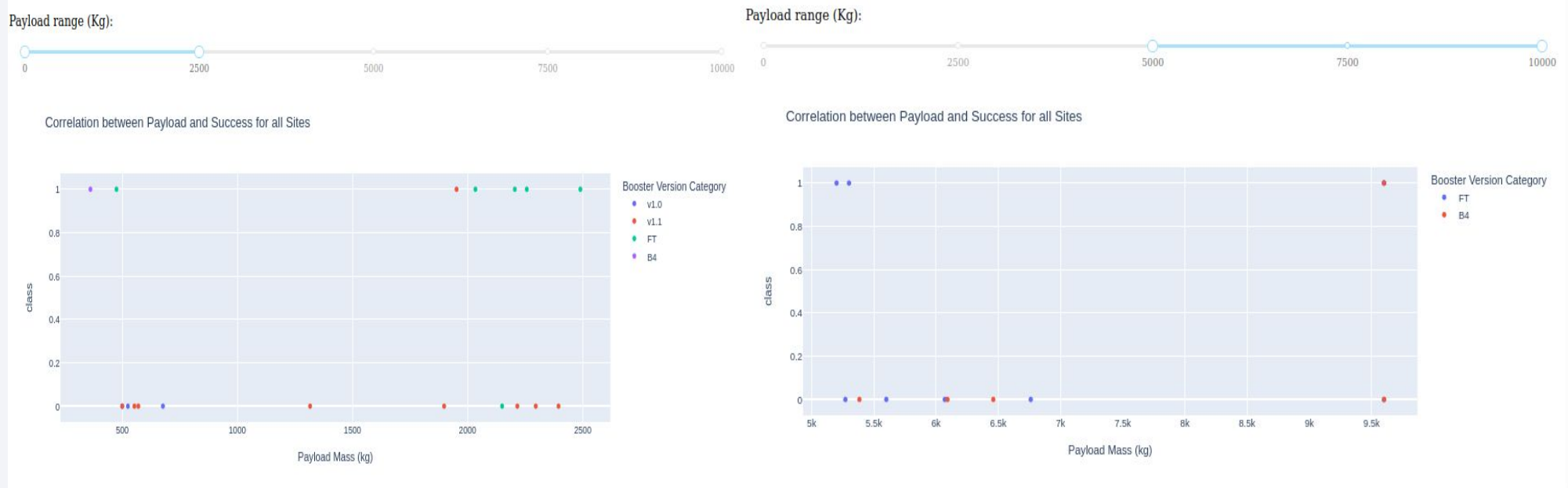
Most Successful Launch Site

KSC LC-39A has 77% successes and 23% failing launches. It is the most successful launch site



• Payload vs. Launch Outcome

- For payloads over 5000 kg it seems that there aren't many successes
- For payloads less than 2000 kg there are plenty of successes (60%), but most of the failures come from booster version **FT** (70% of all the failures)





Section 5

Predictive Analysis (Classification)

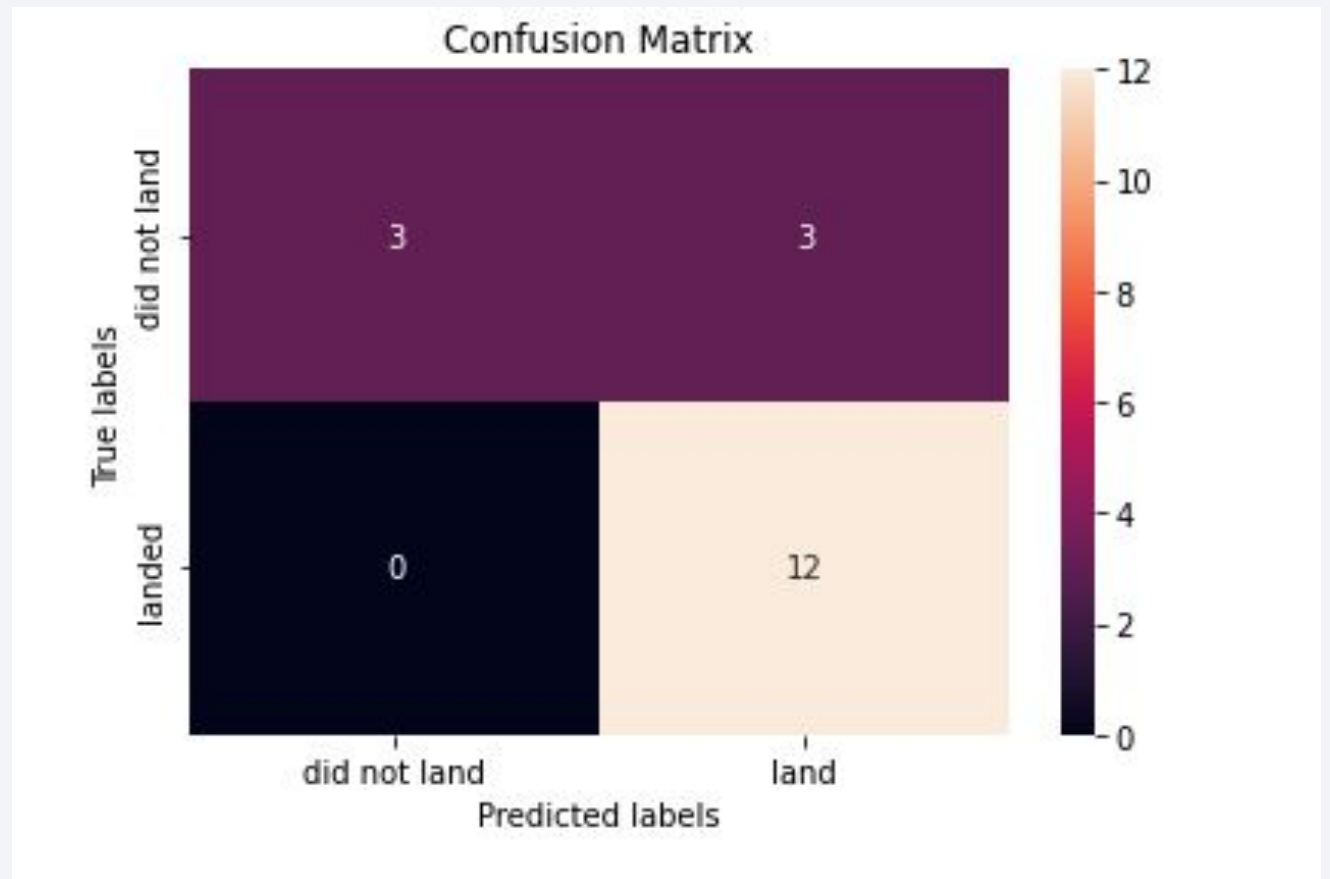
Classification Accuracy

- All of the models had the same accuracy of **83.3%**
- Further metrics can be considered in order to choose the best model

	Methods	Accuracy
0	Logistic Regression	83.33
1	Support Vector Machine	83.33
2	Decision Tree Classifier	83.33
3	K nearest neighbors	83.33

Confusion Matrix

- All the models had the same confusion matrix
- The Specificity is 50%
- The Recall is 100% (the correctly classified % of the rockets which actually land)
- The models predict the rockets that will land but do not do as good a job for the rockets which do not land



Conclusions

- Certain orbital types (**SSO, HEO, GEO, ES_L1**) have **100%** success rate and have therefore less risk and can be priced more freely
- Investigate further why the site KSC LC-39A has the highest number of launch successes
- The model perform well when the rocket actually lands, but there is plenty of room for improvement for the specificity
- The last year there is a lot of improvement of success rate in rocket landing, so the competition in the area might increase
- The payload might play important role in the outcome of the landing for certain booster versions

Appendix

- [GitHub URL](#)

Thank you!

